OP CODE:25900695

MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS)

REGULAR EXAMINATIONOCTOBER 2025

Third Semester

Discipline Specific Elective(DSE)-MG3DSEZIM201

BIOANALYTICAL TECHNIQUES IN MICROBIOLOGY AND IMMUNOLOGY

(2024 ADMISSION)

Duration:2hrs MaximumMarks:70

Part A

Fill in the blanks

Answer all question

Each question carries **1**marks

1. The device inside a centrifuge that holds tubes is called a……(K)/(C01)
Answer:Rotor

2. Thin Layer Chromatography is a type of…….[U]/[CO4)
Answer:Adsorption chromatography

3. Northern blotting is a technique used for the detection of specific …….molecule.[K]/[CO3]
Answer:RNA

4. The part of the antigen recognized by the antibody is called…….[K]/[C06]
Answer:Epitope

5. Laminar air flow cabinets are used in……. laboratories(K)/[C01]
Answer:Microbiology

6. The acid fast staining is used to detect..……. species.[K]/[CO2]
Answer:Mycobacterium

7. In……. PAGE Sodium dodecyl sulfate denatures proteins and gives them a uniform negative charge.[U]/[CO4]
Answer:SDS

8. In western blotting proteins are transferred onto……. .memberane.[U]/[CO3]
Answer:Nitrocellulose

9. Widal test detect antibodies aganist bacteria……..[U]/[CO6].
Answer:Salmonella

10. An……. is any substance that can induce an immune response.[U]/[CO6]

Answer:Antigen [1x10=10]

Part B

Short Answer Questions

Answer any 14 questions

Each question carries 2 Marks

11.**Compare the reliability of pH meter readings in comparison to pH paper[K]/[C01]**
A pH meter is more reliable and accurate than pH paper.

12. Explain why is pencil used in chromatography instead of a pen to draw base line.(U)/[C04]
Pencil is used because graphite is inert and will not dissolve in the solvent

13. Compare RAPD with RFLP in terms of speed and cost.(U)/[C03]
RAPD is faster and cheaper than RFLP.RAPD (Random Amplification of Polymorphic DNA) uses PCR . RFLP (Restriction Fragment Length Polymorphism), which involves restriction digests and probes.

14. Name two types of agglutination reaction.(K)/[C06]
Two types are direct agglutination and passive agglutination.
Direct agglutination involves antigens naturally present on a cell surface, while passive agglutination uses artificial carriers like latex beads coated with antigens or antibodies.

15. Differentiate between simple microscope and compound microscope.(U)/[C02]
A simple microscope uses a single lens, while a compound microscope uses multiple lenses.

16. Explain why basic stains are commonly used for bacterial staining.(U)/[C02]
Basic stains are commonly used because bacterial cell walls are negatively charged.

17. State the principle of a spectrophotometer.(K)/[C04]
The principle is based on the Beer-Lambert law.This law states that the amount of light absorbed by a solution is directly proportional to the concentration of the solute and the path length of the light through the solution.

18. Differentiate between denaturation and extension step in PCR(K)/[C03].
Denaturation separates DNA strands, while extension synthesizes new strands.

19. Define passive immunity. Give one example for naturally acquired passive immunity.(U)/[C06]
Passive immunity is the transfer of active humoral immunity in the form of ready-made antibodies.

20. Explain the principle of ELISA.(U)/[C06]
ELISA (Enzyme-Linked Immunosorbent Assay) uses antibodies and enzyme-substrate reactions to detect antigens or antibodies.

21. Define inoculation.(K)/[C01]
Inoculation is the introduction of a substance into a living organism to produce a disease or immunity to a disease.

22. Define Retention time in chromatography.(U)/[C04]

Retention time is the time taken for a particular solute to pass through the chromatography column under set conditions.

23. Justify the use of AFLP over SSR markers in biodiversity studies.(U)/[C03]

AFLP markers typically reveal more polymorphisms and cover the entire genome, providing higher resolution for biodiversity studies.
SSR markers are co-dominant and highly polymorphic.

24. Write short note on polyvalent vaccine.(U)/[C06]

A polyvalent vaccine is a single vaccine designed to immunize against two or more different strains or types of the same microorganism.

25.Name two surface Markers present on NK cells.(K)/[C05]

Two surface markers present on Natural Killer (NK) cells are CD8 and CD4.

26. Mention any two disadvantage of a phase contrast microscopy.(K)/[C02]

Two disadvantages of phase contrast microscopy are the presence of 'halos' around specimens and the difficulty in observing thick specimens.

[2x14=28]

PartC

Short EssayQuestions

Answer any **8**Questions

Eachquestioncarries**4**marks

27.**Outline the historical development of planar chromatography, how has it involved and impacted on modern chromatography.(A)/[C04] .(Value liberally)**

Planar chromatography has evolved from simple paper chromatography to modern high-performance thin-layer chromatography (HPTLC). Modern advancements like HPTLC offer automated application, development, and detection, providing faster, more sensitive, and quantitative analysis.

28. Explain the application of RFLP in detecting genetic disorders, paternity testing forensic science and other lant breeeding programs.(A)/[C03]

(Any 4 Application -1mark each)

Restriction Fragment Length Polymorphism (RFLP) is used as a genetic marker for identification and diagnosis.For paternity testing and forensic science, it is used for DNA fingerprinting, comparing fragment patterns to establish biological relationships or link suspects to crime scenes. In plant breeding programs, RFLP is used for genetic mapping and marker-assisted selection to improve crop traits.

29. Differentiate between primary and secondary lymphoid organs.(U)/[C05]

(Primary lymphoid organ-2 marks,Secondary lymphoid organ-2marks)

Primary lymphoid organs are where lymphocytes mature, while secondary lymphoid organs are where they are activated.Primary lymphoid organs include the bone marrow (B cell maturation) and the thymus (T cell maturation). Secondary lymphoid organs, such as the spleen, lymph nodes, tonsils, and Peyer's patches, are sites where mature lymphocytes encounter antigens and mount immune responses.

30. Describe the principle and procedure of capsule staining.(U)/[C02]

(Principle-2Marks,Procedur-2 Marks)

Capsule staining is a differential staining method that uses acidic and basic dyes to stain the background and cell, leaving the capsule unstained as a clear halo.The procedure involves mixing bacteria with an acidic stain (e.g., Congo red or nigrosin) and spreading it thinly on a slide. The smear is air-dried (not heat-fixed to prevent capsule shrinkage/destruction) and then counterstained with a basic dye (e.g., crystal violet or safranin) to stain the cell body. The resulting slide shows stained cells surrounded by a clear, unstained capsule against a dark background.

31. Differentiate between temperature regulation in an autoclave and a hot air oven.(K)/[C01]
(Give 2 mark each for temperature and pressure regulation in autoclave and Hot Air Oven)
An autoclave uses moist heat and pressure for temperature regulation, while a hot air oven uses dry heat without pressure.

32. Compare column chromatography with thin layer chromatography.(U)/[C04]

(Principle and Procedure-2Marks each)

Column chromatography is a separation technique performed in a vertical column using a solid stationary phase and a liquid mobile phase, while thin layer chromatography (TLC) is a planar technique using a thin layer of adsorbent on a flat plate.

33. Explain the application of DNA finger printing in various biological fields.(A)/[C03]
(Any 4 application-1 mark each)

DNA fingerprinting is a powerful technique with applications across various biological fields, primarily for identification and determining relatedness.Forensic Science,Paternity Testing: Determines biological parentage with high accuracy,Biodiversity & Conservation: Helps in studying population genetics, identifying species, and tracking illegal wildlife trade.Medical & Agricultural Research: Used for disease diagnosis, identifying pathogens, and in plant and animal breeding programs to select desirable traits.

34. Differentiate between CD4+ and CD 8+ T cells.(K)/[C05]
(Any 4 points each carry 1 marks)
CD4+ T cells are primarily helper T cells that coordinate the immune response, while CD8+ T cells are cytotoxic T cells that directly kill infected or cancerous cells.

35. Write an essay on cells of the immune system.(K)/[C05]
(B cell-2 Marks,T Cells-1 Marks,NK Cells-1 Marks)
The immune system comprises a diverse array of specialized cells that work together to defend the body against pathogens and disease.Lymphocytes (T cells, B cells, NK cells): Responsible for specific recognition of antigens and adaptive immunity. B cells produce antibodies, while T cells mediate cellular responses.Phagocytes (Macrophages, Neutrophils): Engulf and destroy pathogens and cellular debris.Antigen-Presenting Cells (Dendritic cells, Macrophages): Process antigens and present them to T cells to initiate an immune response.Granulocytes (Mast cells, Eosinophils, Basophils): Contain granules with chemicals that mediate inflammation and allergic reactions.

36. Illustrate the working principle of a SEM with a neat labelled diagram.(U)/[C02]
(PRINCIPLE-2 Marks,Diagrammatic representation-2Marks)

A Scanning Electron Microscope (SEM) produces high-resolution images of a sample's surface by scanning it with a focused beam of electrons.Principle: The electron beam interacts with the atoms in the sample, producing various signals like secondary electrons (used for imaging the topography) and backscattered electrons.Procedure: The signals are detected by specialized detectors, and an image is

built up pixel by pixel, showing the surface morphology and composition. A neat, labeled diagram would typically show the electron source, electromagnetic lenses, scanning coils, sample chamber, detectors, and display screen.

[8x4=32]